Lesson 8.5 — Margin of Error

Big idea: A margin of error tells you how much your sample estimate could miss the population value just due to random sampling. It's a "± room for error" around your point estimate.

Quick review

Definition (proportions, 95% confidence): ME ≈ 1.96 · √(p■(1 – p■)/n). Often rounded to 2 · √(…) for quick work.

Reported as: point estimate ± margin of error. e.g., "52% ± 3%".

Effect of n: doubling sample size cuts ME by a factor of √2 ≈ 1.41 (not 2).

Effect of confidence level: higher confidence → bigger margin of error.

Caution: the margin of error covers RANDOM sampling error only. It does NOT correct for bias.

Worked example

Worked example. In a poll of 1,000 voters, 540 support Plan A. Find p■ and the 95% margin of error.

p■ = 0.54. ME = 1.96 √(0.54 · 0.46 / 1000) = 1.96 √(0.000248) ≈ 1.96(0.01577) ≈ 0.031 (about 3.1%).

Warm-ups

1. If p■ = 0.50 and n = 400, estimate ME (95% confidence).
2. If p■ = 0.20 and n = 100, estimate ME.
3. Will a smaller or larger sample produce a smaller margin of error?

Practice

1. A poll of 600 voters has p■ = 0.45. Find ME and the 95% confidence interval.
2. If p■ = 0.30 and you want ME ≤ 0.03 (95% confidence), how large should n be?
3. Two polls report p■ = 0.50 with n = 1,000 and n = 4,000. Compare margins of error.

Challenge

1. A pollster wants to estimate p with ME = 0.02 (95%) and is willing to assume the worst case for p■(1 – p■) (which is 0.25 at p■ = 0.5). Find required n.
2. If a poll says "52% ± 4%", can it confidently claim that the candidate is leading 50%? Why or why not?

Answer key — Lesson 8.5

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $1.96\sqrt{(0.25/400)} = 1.96(0.025) \approx 0.049 \approx 4.9\%$.
2. $1.96\sqrt{(0.16/100)} = 1.96(0.04) = 0.0784 \approx 7.8\%$.
3. Larger sample.

Practice

1. ME = $1.96\sqrt{(0.45 \cdot 0.55 / 600)} \approx 1.96(0.0203) \approx 0.0399$. 95% CI: (0.41, 0.49).
2. $n \geq (1.96^2 \cdot 0.21)/(0.03)^2 \approx (3.8416 \cdot 0.21)/0.0009 \approx 896.4 \rightarrow n \geq 897$.
3. ME at $n = 1000$: $\approx 1.96\sqrt{(0.25/1000)} \approx 0.031$. ME at $n = 4000$: $\approx 1.96\sqrt{(0.25/4000)} \approx 0.0155$. Doubling … wait, n quadrupled, so ME halves ($\approx 1/\sqrt{4}$).

Challenge

1. Worst case p■(1 – p■) = 0.25. $n \geq (1.96^2)(0.25)/(0.02)^2 = (3.8416)(0.25)/0.0004 \approx 2401$.
2. No — 50% is INSIDE the confidence interval (48% to 56%). The poll cannot rule out a tie or close margin. The lead is not statistically significant at 95% confidence.